MIL-E-22200/7B
28 March 1975
SUPERSEDING
MIL-E-0022200/7A(SHIPS)
1 November 1971
MIL-E-15716C
26 December 1961
(See 6.4)

MILITARY SPECIFICATION

ELECTRODES, WELDING, COVERED, MOLYBDENUM

ALLOY STEEL APPLICATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers shielded metal arc electrodes for welding molybdenum alloy steel piping, forgings, and castings.

1.2 Classification. The electrodes shall be furnished in the types, classes and sizes shown in table I as specified (see 6.2).

Table I - Types, classes, and sizes.

Type 1/	Class	Positions of welding	Sizes (diameter) (inch)
MIL-7010-A1	1	All	3/32, 1/8, 5/32, 3/16
	2	Horizontal fillets and flat	7/32, 1/4
MIL-7011-A1	1	All	3/32, 1/8, 5/32, 3/16
	2	Horizontal fillets and flat	7/32, 1/4
MIL-7018-A1	1	All	3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 7/32, 1/4
MIL-7020-A1	2	Horizontal fillets and flat	1/8, 5/32, 3/16, 7/32, 1/4

1/ Type classifications of electrodes will be determined according to the capabilities of the electrodes in meeting the tests specified herein, in the positions of welding specified in table I and in section 4. An electrode will be approved under only one type classification; and an electrode will not be approved if it differs in type from that stated in the manufacturer's request for test.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-10430 - Welding Rods and Electrodes; Preparation for Delivery of.
MIL-E-22200 - Electrodes, Welding, Covered; General Specification for.
MIL-S-22698 - Steel Plate, Carbon, Structural, for Ships.

STANDARDS

MILITARY

MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-418 - Mechanical Tests for Welded Joints.

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PUBLICATION

MILITARY
NAVSHIPS 0900-003-9000 - Radiographic Standards for Production and Repair Welds.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INCORPORATED, AGENT
National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington, DC 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A131-74 - Structural Steel for Ships.
A370-74 - Methods and Definitions for Mechanical Testing of Steel Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
Z49.1-67 - Safety in Welding and Cutting.

(Application for copies should be addressed to the American National Standards Inst Inc., 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Electrodes furnished under this specification shall conform to the requirements of MIL-E-22200 and as specified herein.

3.2 Composition and iron content.

3.2.1 Composition. The chemical composition of the coverings, except as specified in 3.2.2, is optional with the manufacturer. The water content of the coverings shall not exceed the percent-by-weight values specified in table II.

Table II - Water content.

Type	Percent by weight maximum
MIL-7018-A1	1/ 0.40

1/ Not applicable when ordered in class 5 unit containers (see 5.1.2.1).

3.2.2 Total iron content. The total iron content of the covering of MIL-7018-A1 type electrode, including any combined iron in addition to metallic iron powder, shall be not less than 15 percent.

3.3 Groove welds.

3.3.1 Mechanical properties. The mechanical properties of deposited weld metal shall be as specified in table III.

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Table III - Mechanical properties of deposited weld metal, stress-relieved.[1/]

Types	Tensile strength	Yield strength	Elongation in 2 inches (min.)	Charpy-V notch impact properties[2/]	
				Energy (Ft/lbs)	Temperature °F.
	P.s.i. (Min.)	P.s.i. (Min.)	Percent	Minimum average	Minus
All	70,000	57,000	22	20	20

1/ Stress-relieved in accordance with note 6 or note 8 of figure 1.

2/ Impact properties shall be determined for qualification and comparison inspection only.

3.4 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table IV.

Table IV - Chemical composition (percent) of deposited weld metal.[1/]

Types	Carbon	Manganese	Silicon	Sulphur	Phosphorus	Molybdenum
MIL-7010-A1	0.12	0.30-0.60	0.40	0.030	0.035	0.40-0.65
MIL-7011-A1	0.12	0.30-0.60	0.40	0.030	0.035	0.40-0.65
MIL-7018-A1	0.12	0.50-0.90	0.30-0.80	0.030	0.030	0.40-0.65
MIL-7020-A1	0.12	0.30-0.60	0.40	0.040	0.040	0.40-0.65

1/ Single values are maximums.

3.5 Electrode identification. Electrodes shall be identified as specified in 3.5.10 of MIL-E-22200.

3.6 Electrode lengths. The electrode lengths shall be as specified in table V (see 6.2).

Table V - Electrode lengths.

Types	Sizes	Electrode lengths end-grip
	Inch	Inches
MIL-7010-A1	3/32	9, 12, or 14
MIL-7011-A1	1/8, 5/32	9 or 14
MIL-7018-A1	3/16	14
	7/32, 1/4	18
MIL-7020-A1	1/8	14
	3/16	18
	7/32, 1/4	18

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-E-22200 and as specified herein.

4.2 Qualification tests. Qualification tests shall be as specified in 4.2 of MIL-E-22200 and tables IX and XI.

4.3 Comparison inspection. In addition to the examination and tests required in 4.5 of MIL-E-22200, tests specified in table VI shall be conducted as specified in 4.3 of MIL-E-22200, when required by the Naval Ship Engineering Center (NAVSEC).

4.4 Quality conformance inspection.

4.4.1 Lot. A lot of electrodes shall be as specified in 4.4.4 and 4.4.6 of MIL-E-22200 for inspection level C.

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Table VI - Comparison inspection.

Types	Sizes	Tests	Requirements
All	All	4.6.8 (MIL-E-22200)	3.4 and table IV herein
All	All	Weld test No. 3 (see table IX herein)	3.3.1 and table III herein

4.4.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be in accordance with 4.4.7 through 4.4.8.3 of MIL-E-22200 for tests specified in MIL-E-22200 where required, and as specified in 4.4.3 for quality conformance weld tests.

4.4.3 Quality conformance inspection. Quality conformance tests shall be conducted on samples selected as specified in 4.4.2. One sample representing one wet mix of covering mixture in each lot shall be subjected to all the tests required by table VII (i.e., X-ray, mechanical and complete chemical analysis). Samples representing each of the remaining wet mixes of covering mixture in a lot shall be tested as specified in table VIII to assure equivalency as required by 4.4.4.2 of MIL-E-22200.

Table VII - Quality conformance inspection.

Types	Tests	Procedures
All	Test No. 3 and 8a[1/] Table XI	Table IX
All	Chemical analysis	Test No. 3, table IX

1/ Impact properties shall be determined only when required by the procurement document (see 6.2).

Table VIII - Wet mix equivalency tests.

Types	Tests	Procedures
All	Assure equivalency[1/]	4.4.4.2 of MIL-E-22200

1/ See 4.4.4.2 of MIL-E-22200.

4.4.4 Unsatisfactory test results. If the results of the first tests are determined to be unsatisfactory, two retests shall be permitted. The results of both retests shall be satisfactory for quality conformance inspection.

4.5 Test procedures.

4.5.1 Weld tests. Table IX summarizes the weld tests required under this specification for qualification testing of electrodes and for the inspection as specified in tables VI, VII, and VIII. Table X shows base plate material required for each particular electrode type under that test.

4.5.2 Other tests. Table XI summarizes tests, in addition to those specified in table IX, required for qualification testing of electrodes for the QPL, and for quality conformance inspection, as applicable.

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Table IX - Summary of required weld tests.[1/]

Weld test No.	Types	Diameter sizes						Weld joint design procedure	Welding position[2/]	Tests				Chemical or metal-lurgical	Requirements
		3/32	1/8	5/32	3/16	7/32	1/4			X-ray	Tensile[4/]	Impact	Bend		
3	All	X	X	X	X	X	X	Figure 1 and notes	F	[3/]	X	X	--	Chemical analysis[5/]	3.7.1 of MIL-E-22200 and tables II and III herein
8a	MIL-7018-A1	X ---	X ---	X ---	--- X	--- X	--- X	Figure 2 and notes	V F	X[3/] X[3/]	--- ---	--- ---	-- --	---- ----	3.7.1 of MIL-E-22200
8a	MIL-7010-A1 MIL-7011-A1 MIL-7020-A1	X ---	X ---	X ---	X ---	--- X	--- X	Figure 2 and notes	V F	X[3/] X[3/]	--- ---	--- ---	-- --	---- ----	3.7.1 of MIL-E-22200

[1/] For qualification testing of the electrodes for the QPL, all electrode types shall be tested using direct current (d.c.) reverse polarity (r.p.). In addition, MIL-7011-A1 and MIL-7018-A1 electrode types shall be tested using alternating current (a.c.) and MIL-7020-A1 electrode type shall be tested using d.c. straight polarity (s.p.) and a.c. For comparison inspection, when required, and quality conformance inspection, all electrode types shall be tested using d.c. r.p.

[2/] F = flat, V = vertical.

[3/] When both tests (3 and 8a) are applicable, radiography of test No. 3 will satisfy the requirements for test No. 8a for class 2 electrodes. If test No. 3 is radiographed, test No. 8a may be eliminated for class 2 electrodes.

[4/] Includes ultimate, yield, elongation, reduction in area.

[5/] May be determined from milling sample taken from ruptured ends of tensile specimens or drilling. In event of variances with composition requirements or conflicting results by testing activities, analysis shall be verified by procedures specified in 4.6.8 of MIL-E-22200.

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Table X - Weld joint base plate.

Types	Base plate
All	Medium steel MIL-S-22698 or ASTM A131-74

Table XI - Summary of other tests.

Tests	Procedures	Requirements
	MIL-E-22200	
Visual and dimensional	4.6.2.3	3.10 (MIL-E-22200 and 3.6 herein)
Concentricity	4.6.2	3.5.2 (MIL-E-22200)
Dielectric strength 1/	4.6.3	3.5.4 (MIL-E-22200)
Flaking and cracking of covering 1/	4.6.4	3.5.5 (MIL-E-22200)
Covering moisture	4.6.9	3.2.1 herein
Total iron content in covering 1/	4.6.10	3.2.2 herein

1/ Required for qualification inspection only.

4.6 Certificate of test. A certificate of quality conformance of the weld tests specified in 4.4.3 (i.e., X-ray, mechanical, and chemical analysis), containing actual quantitative test and inspection results including the range of amperage employed in making the required welds for quality conformance inspection for each accepted lot included in a particular shipment shall be furnished to the consignee with the shipment. The minimum quality conformance test result data required shall be as shown on figure 3 and shall be submitted on the form shown.

4.7 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Electrodes shall be packaged in class 1a or class 1c unit containers, except that when class 1b unit containers are specified by the contract or order (see 6.2), 12-, 14-, and 18-inch electrodes may be supplied in 50-pound unit containers.

5.1.1.1 Special instructions. When electrodes are packaged in class 1a unit containers in accordance with MIL-W-10430, fiberboard or other cushioning material shall be inserted in each end of the unit container to restrict end-wise movement of the electrodes. The cushioning materials shall be dry and in equilibrium with the electrode covering. Class 1a containers shall be of a snug fitting design with the container length not to exceed the rod length by 1/2-inch.

5.1.2 Level C. Packaging shall afford protection against deterioration and physical damage during shipment from the supply source to the using activity. The packaging applied shall insure conformance to the stability requirements specified in 3.5.8 of MIL-E-22200.

5.1.2.1 When specified (see 6.2), type MIL-7018-A1 electrodes may be packaged in class 5 unit containers in accordance with MIL-W-10430.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A and B. Electrodes packaged as specified (see 6.2), shall be packed level A or B in accordance with MIL-W-10430. Palletization, when required (see 6.2), shall be in accordance with MIL-W-10430.

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5.2.2 Level C. Electrodes, packaged as specified (see 6.2), shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or Regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required herein or by the contract or order (see 6.2), interior packages and exterior shipping containers and palletized loads shall be marked in accordance with MIL-W-10430.

5.3.1 Special marking.

5.3.1.1 Interior packages and shipping containers. In addition to the lot identification required by 4.4.6 of MIL-E-22200, identification on interior packages shall include MIL-type, size, specification number, manufacturer's or distributor's name, date of manufacture (month/year), and manufacturer's or distributor's brand or type designation. In addition, all unit packages shall be marked or labeled with the following precautionary information, as a minimum and prominently displayed:

CAUTION
Welding may produce fumes and gases hazardous to health. Avoid breathing these fumes and gases. Use adequate ventilation. See ANSI Z49.1 "Safety in Welding and Cutting" published by the American Welding Society.

Significant toxic constituents when present in the electrode in greater than trace amounts shall be included in the precautionary information so that normal ventilation can be increased accordingly. These constituents include but are not limited to those itemized in section 8.5 through 8.11 of ANSI Z49.1-67.

5.3.1.2 Class lb containers. Each class lb container on the container end adjacent to the electrode grip end shall be marked "OPEN THIS END".

6. NOTES

6.1 Intended use. Electrodes covered by this specification are intended for use in welded fabrication of piping, forgings, and castings made by molybdenum alloy steel.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and size required (see 1.2).
(c) For the 3/32, 1/8, and 5/32 size electrode, the length required (see table V).
(d) Whether quality conformance inpact tests are required (see table VII and footnote).
(e) Levels of packaging and packing required (see 5.1 and 5.2).
(f) When class lb unit containers are required (see 5.1.1).
(g) Whether type MIL-7018-A1 electrodes should be packaged in class 5 unit containers (see 5.1.2.1).
(h) When palletization is required (see 5.2.1).
(i) Special marking required (see 5.3).

6.2.1 Electrodes should be ordered by the pound.

6.3 Inspection after delivery. Post delivery inspection of electrodes to determine conformity to this specification and for acceptance thereof is the responsibility of the consignee.

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6.4 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

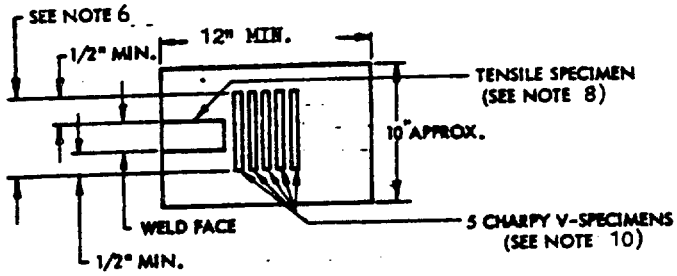
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Preparing activity:
Navy - SH
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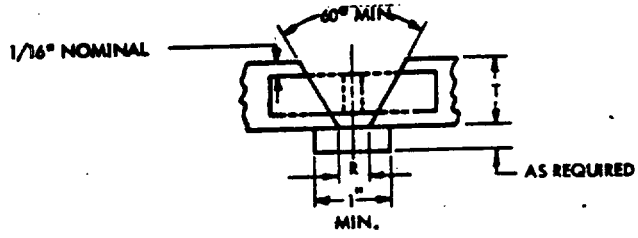
Review activities:
Navy - SH
Air Force - 11

User activities:
Army - AT
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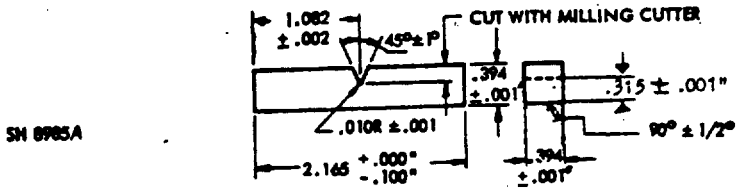


LAYOUT OF TEST PLATE



GROOVE PREPARATION OF TEST PLATE

SHOWING ORIENTATION OF CHARPY-V TEST SPECIMEN



CUT NOTCH WITH MILLING CUTTER. MAKE FOUR CUTS PER NOTCH IN FOLLOWING ORDER: a) .040; b) .025; c) .010; d) .004. NO IRREGULARITIES IN CONTOUR OF APEX OF NOTCH. NO. 8 POLISH MICRO-INCH FINISH.

Figure 1 - Welded joint for all-weld-metal tension test and impact test (test No. 3).

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Notes to figure 1:

1. Base metal shall be in accordance with table X.
2. Impact tests are required for qualification (QPL) and for comparison inspection.
3. Welds shall be deposited in the flat position using d.c. and a.c. as required by note 5 to table IX. The actual amperage values used shall be within the manufacturer's recommended current range and shall be recorded and reported. The root weld layer may be deposited with 1/8-inch diameter electrodes when testing 3/16-inch and larger diameter electrodes, provided both electrodes have the same manufacturer's designation. Welds shall be deposited in approximately 1/8-inch thick layers using a 2-1/2 core wire diameter maximum weave buildup sequence. The final layer shall include a reinforcement of standard proportions (1/32-inch to 1/8-inch). Each layer shall be started at the finishing end of the preceding layer. The operator error, pass and layer sequence shall be recorded and reported as specified in table XII.
4. After the joint has been tacked, the assembly shall be heated to 225° ± 25°F prior to welding. After completion of each layer the assembly shall be allowed to cool in still air at room temperature and the subsequent layer shall be started where the temperature of the plate has dropped to 225° ± 25°F. If it is necessary to interrupt this procedure, the assembly shall be preheated to 225° ± 25°F.
5. Radiographic examination of test No. 3 is not required, but may be used in lieu of radiography of test No. 8a for class 2 electrodes when test No. 3 is radiographed. After completion of welding the weld reinforcement and the backing shall be removed flush with the base plate surfaces. The assembly shall be radiographed in accordance with MIL-STD-271. Radiographs shall meet acceptance standards of NAVSHIPS 0900-003-9000 as specified in 3.7.1 of MIL-E-22200.
6. After the completion of the welding or after radiography, if used, the test weldment shall be cut into two coupons, one for the all-weld-metal tensile specimen and one for the five charpy V-notch impact specimens as shown. No base metal shall be removed within 1/2-inch of the edges of the face of the weld by flame cutting. Only sawing or machining shall be used within 1/2-inch of the weld. Note: Impact tests are not required for quality conformance tests unless specified in the procurement document (see tables VII, VIII and IX).
7. The tensile coupon, prior to machining, shall be stress-relieved using the following procedure:

   (a) Employing an electric furnace the tensile test coupon shall be heated to 1150° ± 25°F, held for 2 hours at temperature and allowed to furnace cool at a rate not to exceed 200°F per hour maximum to a temperature below 500°F.

8. The stress-relieved tensile coupon shall be etched, when necessary, to establish the location of the weld. The coupon shall be machined into an all-weld-metal tensile specimen, centered in the weld, 0.505-inch diameter specimen for 3/4-inch thick weldment and 0.357-inch diameter specimen for 1/2-inch thick weldment. See MIL-STD-418 or ASTM A370-74 for details. The specimen shall be tested for conformance with table III herein.
9. The impact test coupon, when required, shall be subjected to the following stress-relieving conditions:

   (a) Using an electric furnace, the impact test coupon shall be held for 2 hours at 1150° ± 25°F and then allowed to cool to a temperature below 500°F at a rate not to exceed 200°F per hour.

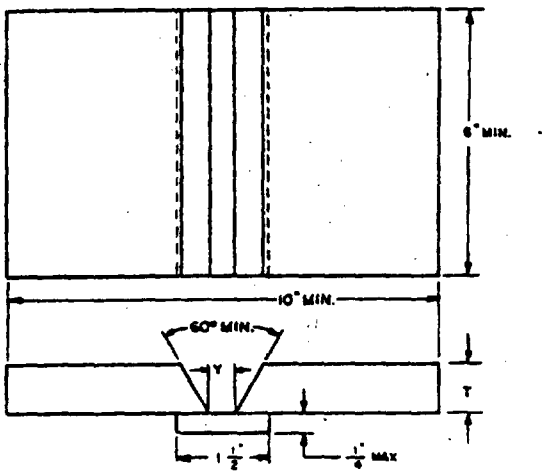
10. The stress-relieved impact test coupon shall be etched, when necessary, to establish the location of the weld. Five charpy V-notch specimens shall be machined in accordance with details shown on figure 1.
11. Impact properties for the five specimens shall be obtained at test temperatures specified in table III herein, plus or minus 3°F. The extreme lowest impact value thus obtained together with the extreme highest value may be disregarded for this test. Two of the three remaining values shall be greater than the specified 20 foot/pounds energy level; one of the three may be lower but shall be not less than 15 foot/pounds; the computed average value of these three shall be greater than the specified 20 foot/pound level (see table III herein).

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Table XII - Groove weld preparation.

Electrode size	Minimum thickness (T)	Minimum root opening (R)	Number of layers	
			Min.	Max.
(Inch)	(Inch)	(Inch)		
3/32	1/2	1/4	1/	1/
1/8	1/2	1/4	1/	1/
5/32	3/4	1/2	7	9
3/16	3/4	1/2	7	9
7/32	3/4	1/2	7	8
1/4	3/4	1/2	9	11

1/ Number of layers shall be reported.



Electrode size	3/32 (Inch)	1/8 (Inch)	5/32 (Inch)	3/16-1/4 (Inch)
Thickness (T) 1/	1/4	3/8	1/2	1/2
Root opening (Y) 2/	1/4	5/16	3/8	1/2
Weld bead width (max.)	3/8	1/2	5/8	3/4

1/ Minimum.
2/ Tolerance ± 1/16-inch.

Figure 2 - Groove weld usability test (test No. 8a).

Notes to figure 2:

1. Welding shall be accomplished with the test plate in the vertical or flat position as specified in table IX using d.c. and a.c. as required by note 5 to table IX. The actual amperage values used shall be within the manufacturer's recommended current range and shall be recorded and reported.
2. Base metal shall be in accordance with table X.
3. The edge preparation shall be flame cut and ground to remove slag and surface irregularities.
4. Welds shall be deposited in approximately 1/8-inch thick layers. Weld bead widths shall be as shown above. Each bead shall contain a start in the area to be evaluated.
5. The reinforcement shall be 3/16-inch maximum.
6. Preheat and interpass temperature shall be 225° ± 25°F.
7. After completion of the weld, the weld reinforcement and backing strips shall be removed flush with the base plate on both surfaces and the assembly shall be X-rayed in accordance with MIL-STD-271.
8. The radiograph shall meet the X-ray standards of NAVSHIPS 0900-003-9000 as specified in 3.7.1 of MIL-E-22200.

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CERTIFICATION OF QUALITY CONFORMANCE TESTS

Manufacturer or Distributor _____ Customer's Name _____

Address _____

Date _____ Customer's Order No. _____

Specification MIL- _____ Core Wire Heat No. _____

Type MIL- _____ Lot Identification MIL-E-22200 Para. _____

Diameter & Length _____

Inspection Level _____ Wet Mix No. _____

Lot No. _____

Chemical Analysis (Complete)

Carbon _____

Manganese _____

Silicon _____

Sulfur _____

Phosphorus _____

Molybdenum _____

Mechanical Test Stress-Relieved

Yield Strength (0.2% offset method) _____

Tensile Strength _____

Elongation (%) _____

Reduction of Area (%) _____

Chemistry was taken from: ☐ Chem Pad ☐ Groove Weld

X-Ray: _____

Concentricity (%) _____

Covering Moisture _____

Covering Iron Content _____

Groove Weld Test

Test No.	3	8a	Chem Pad
Amperage	_____	_____	_____

Grinding During 8a Test Plate Preparation (see note 3) Operator Error (Layer Nos.) _____

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____

NOTES:

1. Items not applicable to the type electrode involved or exempted by the procurement document shall be marked N/A.
2. This form is applicable to the quality conformance tests required for each lot. A copy of the quality conformance test certification for the lot (figure 3) shall be forwarded with each shipment.
3. See 3.5.7 and 3.9 of MIL-E-22200.

Figure 3 - Certification of quality conformance tests.

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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